# 1 Basics/Revision

- $i = \sqrt{-1}$ is called the imaginary/complex number
- $\mathbb{C}$ is the set of all numbers of the form $z = x + yi$, i.e $\mathbb{C} = \{z : z = x + yi, x, y \in \mathbb{R}\}$
  - $x$ is the real part of the complex number, i.e $\Re(z) = x$
  - $y$ is the imaginary part of the complex number, i.e $\Im(z) = y$
- The magnitude/norm over complex numbers $z = x + yi$ is defined as $|z| = \sqrt{x^2 + y^2}$
- Conjugate of a complex number $z = x + yi$ is defined as $\overline{z} = x - yi$
  - $z \cdot \overline{z} = |z|^2$
- $\arg(z) = \theta \in (-\pi, \pi]$ where $\theta$ is the angle between the positive real axis and the line segment joining the origin to the point $(x, y)$ in the complex plane.
- **adding and subtracting complex numbers**
- **multiplying complex numbers**
- **dividing complex numbers**
- Geometric interpretation of $|z_1 - z_2|$?
- **Polar form** of a complex number $z = x + yi$ is given by $z = rcis(\theta) = r(\cos\theta + i\sin\theta)$= $\mathbf{re^{i\theta}}$ where $r = |z|$ and $\theta = \arg(z)$

- Multiplying/dividing with complex numbers is easier in polar form
- **Mathematica**
  - `Abs[z]` gives the magnitude
  - `Arg[z]` gives the argument
  - Can also use `ArcTan[x,y]`!!
  - `Conjugate[z]` gives the conjugate
  - `Re[z]` ,`Im[z]` gives the real and imaginary parts
  - `ComplexExpand[z]` gives the cartesian form of a complex number
  - Will send mathematica code for more functions `:)`

1. For $z_1 = 3 + 4i$, find $|z_1|, \overline{z_1}, \arg(z_1)$ and express $z_1$ in polar form.

2. Let $z_2 = 2cis(\pi/4)$. Find $z_2$ in cartesian form.

3. Find the product and quotient of $z_1$ and $z_2$ in both cartesian and polar forms.

- **De Moivre's Theorem** for any complex number $z = rcis(\theta)$ and any integer $n$, we have $z^n = r^n cis(n\theta)$: Use to calculate powers of complex numbers
- **Roots of complex numbers** If $z = rcis(\theta)$, then the $n$-th roots of $z$ are given by $z_k = r^{1/n} cis\left(\frac{\theta+2k\pi}{n}\right)$ for $k = 0, 1, ..., n-1$
- Geometric interpretation of the roots of a complex number?

# 2 Complex polynomials

- $\mathbb{C}$ is algebraically closed, i.e every non-constant polynomial with complex coefficients has at least one complex root.
- This means that a polynomial of degree $n$ has exactly $n$ roots in $\mathbb{C}$ (counting multiplicities)! (How?) i.e **Fundamental theorem of algebra**
- **Conjugate root theorem** If a polynomial has real coefficients and a complex root $z$, then its conjugate $\overline{z}$ is also a root.
- Vieta's formulas for a polynomial of degree 3 with roots $r_1, r_2, r_3$ and leading coefficient $a_3$:
  1. $r_1 + r_2 + r_3 = -\frac{a_2}{a_3}$
  2. $r_1 r_2 + r_1 r_3 + r_2 r_3 = \frac{a_1}{a_3}$
  3. $r_1 r_2 r_3 = (-1)^3 \frac{a_0}{a_3}$